🔐 Kubernetes Lab Challenge: Inter-Pod Communication with NetworkPolicy

🎯 Objective
Design a microservice-style setup in your EKS cluster where multiple pods interact, and restrict traffic between them using **NetworkPolicies**.

🧭 Challenge Description

You are tasked with simulating a small system composed of:

- A **frontend pod** (e.g., a simple web server or static UI)
- A **backend pod** (e.g., an API or data processor)
- A **third pod** (e.g., an unauthorized scanner or test client)

Then apply **NetworkPolicies** to enforce the following:
- ✅ Frontend can access Backend
- 🚫 Third pod **cannot** access Backend
- ✅ Kubernetes DNS resolution still works

💡 What You’ll Learn

- Kubernetes Services and DNS-based Pod-to-Pod communication
- How to isolate pod traffic using labels and selectors
- Designing least-privilege communication paths with NetworkPolicy

---

🔧 Tasks

🟢 Step 1: Define Pod Deployments
- Create 3 pod definitions:
  - `frontend` pod (any HTTP client or UI)
  - `backend` pod (simple API or dummy server)
  - `scanner` pod (just a busybox or curl-based pod)

📘 Reference:
- https://kubernetes.io/docs/concepts/workloads/pods/

🟢 Step 2: Create a Service for Backend
- Expose the backend pod via a Kubernetes Service (ClusterIP).

📘 Reference:
- https://kubernetes.io/docs/concepts/services-networking/service/

🟢 Step 3: Write a NetworkPolicy
- Allow only `frontend` to talk to the backend (using pod labels)
- Deny access from all other pods (default deny behavior)

📘 Reference:
- https://kubernetes.io/docs/concepts/services-networking/network-policies/

🟢 Step 4: Test Communication
- Use curl or wget from the `frontend` and `scanner` pods:
  ```bash
  curl http://<backend-service-name>
  ```

Expected behavior:
- ✅ `frontend` can access the backend
- 🚫 `scanner` cannot access the backend

✅ Completion Criteria

- All 3 pods are deployed
- NetworkPolicy correctly restricts traffic
- `kubectl get networkpolicy` shows your policy
- You verified outcomes via logs or terminal

📦 Bonus Tasks (Optional)

- Introduce namespace-based isolation
- Allow only specific ports (e.g., port 80 or 443)
- Create an audit log pod using a sidecar

🚀 Submission Instructions

Be ready to:
- Explain your NetworkPolicy
- Show test results via terminal or screenshot

```
  - Share your manifest files (optional)
```